

Deep Learning Framework for Crowd Density Estimation and Motion Analysis

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Abstract—The big crowds at events such as religious festivals, political meetings, concerts, and sports events in places like India often result in overcrowding of the places, which are prone to erupting into stampedes, leading to serious injuries and loss of lives. In this paper, we introduce an AI-based crowd monitoring system that processes live CCTV footage to determine the density of the crowd on a per-spatial area scale, determine movement patterns in real-time, and provide a proactive alert to a user via an interactive dashboard. It uses YOLOv11 to detect the people with high speed and accuracy and combines it with the Lucas-Kanade optical flow algorithm to detect abnormal crowd surges. Seaborn and Matplotlib are used to render density distributions in terms of temporal analysis charts and heatmaps, whereas an analysis control panel is presented that displays the main insights such as the maximum crowd count, the highest density, the peak motion surge, and risk distribution. A line chart linked to a threshold also helps the operators to differentiate between safe levels and high levels of crowds. The system was able to reach a precision of 0.89, a recall of 0.86, and an mAP@0.5 of 0.83, which indicated good detection capability. This framework has the potential to help manage large-scale events by providing a scalable and cost-effective answer to the management of public safety by fully automating the monitoring pipeline and allowing it to operate without relying on manual intervention.

Keywords—Crowd density, Behavioural patterns, stampede risk, real-time detection, Optical Flow (Lucas–Kanade Method), Seaborn, Matplotlib, heatmaps, YOLO v11,

I. INTRODUCTION

In some countries such as India, public gatherings are very entrenched in their daily lives. Their cultural unity and group identity are expressed through religious festivals, sport events, political meetings, and national holidays, which attract millions of participants on a regular basis. But the very size of such events puts forward serious safety

concerns, the primary of which is the danger of stampedes. The overpopulation as well as insufficient crowd control may turn festive areas into life-threatening areas quickly, causing serious injuries and deaths. The current surveillance systems, which are based mainly on the CCTV footage that is checked by human operators and have little police presence on the ground, have always been too slow and reactive to be able to avert such disasters on time [1]. The current developments in the fields of artificial intelligence and computer vision provide an opportunity for a paradigm shift to overcome these failures. With the direct integration of the deep learning-based models of the detection into the already existing CCTV infrastructure, the continuous, precise, and large-scale monitoring of the crowd conditions is possible without any substantial costs in terms of hardware. The present paper suggests a crowd monitoring system, which will utilize AI to detect people in real-time using YOLOv11 and track movements using the Lucas-Kanade optical flow algorithm, thus allowing the system to identify surges of abnormal crowds before they grow. The density distributions are presented in the form of a temporal heatmap and interactive analysis dashboard, revealing some of the major metrics such as the maximum crowd count, maximum density, and the values of motion surge. The proposed system will allow the authorities to proactively respond to the full monitoring pipeline, thus involving less reliance on human intervention and assessing the outcomes of the implementation by thousands of people with high effectiveness in the context of large-scale events [2].

II. LITERATURE SURVEY

As new methods have emerged, crowd analysis has become better and better over time. Since their discovery, deep learning algorithms have found several uses, thanks

to their selective power and quick functional extraction. Present methods for monitoring and analyzing dense crowds can be hindered by this different trait. Researchers in the field of crowd analysis should come up with innovative methods to address the growing requirement for computer vision in real-time surveillance video monitoring and analysis of large groups of people [3][4]. Object detection is a computer vision based method that finds examples of a specific class of semantic items in video and picture files. Among the many typical uses of object detection is crowd density analysis. Convolutional neural network (CNN) models are helpful for crowd density classification techniques because they overcome obstacles such as non-uniform density, occlusion, and variances across and within scenes. This study introduces a model for video surveillance systems called MDTL-ICDDC, which uses metaheuristics and deep transfer learning to detect and classify crowd densities [5].

The paper [6] possesses a big mesh-grid number, a fine-grained mesh size, and a high user sample, which are some of the advantages our dataset has over the previous ones. We present DeepCrowd, a new deep learning model that utilises pyramid architectures and a Convolutional LSTM-based high-dimensional attention mechanism, to tackle this massive crowd dataset. In the end, we show that the suggested DeepCrowd is better than many state-of-the-art approaches by conducting extensive performance evaluations. Wisual is a device-free framework based on CSI, which improves the estimation of the density of crowds by allowing the fine-grained distinction between the movement trajectories of multiple individuals with FoC and JMFP features. It uses a CS-3DCNN framework to perform high-quality multi-target analysis, and Wi-Fi imaging via 2D-MUSIC is a framework that offers a low-quality visualization of the space. The methodology eliminates the constraints of the conventional IoT techniques by enhancing the path-level resolution and shows robustness and scalability in the environment of dynamic indoors with up to 98 percent accuracy [7].

A framework for collective motion analysis models individual spatio-temporal behavior using an intention-aware approach and structure-based collectiveness measure. A multi-stage clustering method detects local and global motion patterns, achieving improved performance and supporting applications like anomaly detection [8]. This paper [9] suggests a ConvLSTM model and an encoder-decoder network as it attempts to estimate crowd density and flow based on sequential data to predict in real time the dangerous patterns of crowds. The system is tested against actual data and provides the authorities with the opportunity to act proactively, enhancing the safety of the population in large cities by allowing them to implement timely interventions during high-density crowd situations.

This study [10] proposes CDEM-M, a crowd density estimation and mapping method that combines surveillance video with GIS, using semantic segmentation and a BP neural network to accurately estimate crowd density in large-scale, high-altitude scenes. The system achieved a segmentation accuracy of 96.70% and significantly outperformed density map-based methods, reducing MAE and RMSE by 89.9 and 85.1 respectively. In this paper [11], DCNNCDM-IUP, a smart city framework of crowd density monitoring integrating SE-DenseNet, red fox optimization, and ConvLSTM, is presented, with the result of 98.40% accuracy relying on diverse crowds. Although it has been working well, the model has a fixed learning rate, and it cannot be scaled on large datasets, which raises the issue of more adaptive, real-time methods of crowd monitoring that could improve its performance and applicability in dynamic environments. This study [12] evaluates three YOLO-based crowd counting models, finding that YOLOv3 with Canny Edge Detection achieved the best performance with 86.8% accuracy and 75% precision. While promising, these results fall short compared to newer architectures, reinforcing the need for more advanced models.

A. Limitations of Existing Methods

Despite significant progress in crowd density estimation, several limitations persist across existing approaches. Models such as CSRNet and MCNN achieve high accuracy but demand powerful GPUs, making real-time deployment computationally expensive, while the DCNNCDM-IUP framework's fixed learning rates limit adaptability to dynamic crowd conditions. Detection-based models like YOLOv3 and YOLOv7-tiny offer fast inference but struggle with missed or duplicate detections in extreme densities, and motion analysis methods like Lucas-Kanade optical flow break down in large, chaotic crowds. GIS-based systems such as CDEM-M are restricted to high-altitude setups, failing to generalize across diverse surveillance environments. Most critically, existing studies lack an integrated end-to-end framework combining real-time detection, motion analysis, density visualization, and automated alerting — a gap that the proposed system directly addresses.

III. SYSTEM ARCHITECTURE

Figure 1 shows the system is designed according to three functional layers. The input and detection layer processes raw video captured by CCTV cameras and preprocesses it by extracting frames and resizing them and removing the background prior to transmitting it to YOLOv11 to perform real-time person detection, which generates bounding boxes and a people count per frame. The second layer does parallel behavioral analysis; crowd density estimation classifies areas as normal or high-risk based on pixel

distribution; and Lucas-Kanade optical flow identifies the movement of individual objects to identify sudden changes in velocity or direction, which is in turn a sign of a surge or stampede. The third layer produces output by rendering density results as heatmaps using Seaborn and Matplotlib, which are then aggregated on a surveillance dashboard; any identified risk condition will trigger an alert to security personnel, law enforcement agencies, and entry gates [13-15].

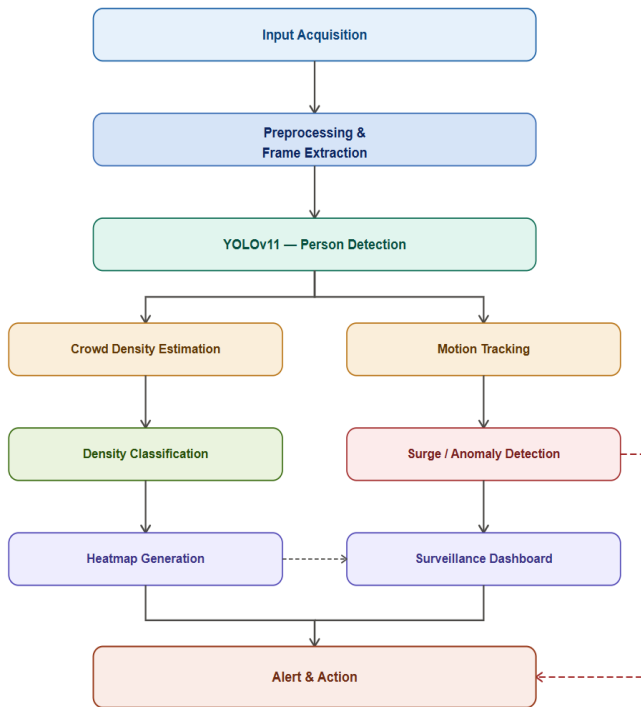


Fig.1. System Architecture

The system is structured into six sequential layers in the framework as in Fig 2. The Data Acquisition Layer receives raw video from CCTV cameras or recorded sources. The preprocessing and frame extraction layer is used to clean and preprocess frames.

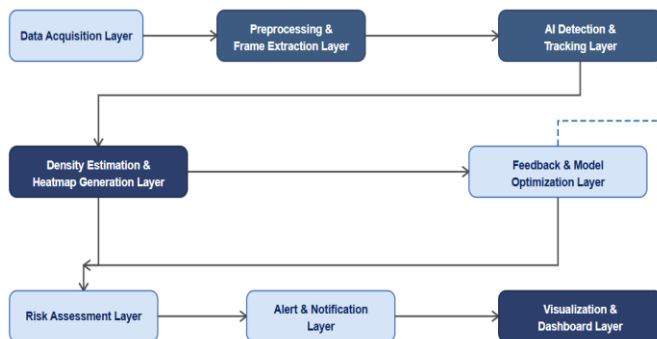


Fig.2. Framework of the proposed system

The AI Detection and Tracking Layer uses person detection with the help of YOLOv11 and motion tracking

by the use of Lucas-Kanade optical flow. The Density Estimation and Heatmap Generation Layer is used to calculate the density of crowds and generate colored heatmaps. Feedback and Model Optimization Layer The detection model is constantly improved by a feedback and model optimization layer, using feedback of the previous model output. Lastly, the Risk Assessment Layer assesses the levels of threat and forwards them to the Alert and Notification Layer, and all findings are shown on the Visualization and Dashboard Layer to be monitored by security staff in real-time.

IV. METHODOLOGY

A. Data Collection and Preprocessing

The system is learned on the ShanghaiTech crowd counting dataset, which is made of 1,998 annotated images, and the number of human head annotations is about 330,165 obtained in the real-world setting, such as stadiums, streets, and events. Crowd benchmark datasets that are publicly available are used in the case where live CCTV footage is not available. Preprocessing provides consistency to all inputs—video frames are scaled to a regular resolution, filters designed to remove noise are used to make the image clearer, and normalization is used to remove lighting differences between day and night. Frame stabilization is used before optical flow calculations in order to eliminate camera shake effects, and cropping and contrast normalization of facial areas are done to ensure accurate detection of expressions in low-light situations.

B. Feature Extraction and Selection

Video, facial, and textual data modalities are extracted to identify critical features that provide maximum predictive accuracy without consuming excessive computational costs. The features that can be extracted out of the video frames are per-frame counts of individual counts, pixel-level spatial positions, and movement vectors calculated by the Lucas-Kanade optical flow technique, which make it possible to quantify motion trajectories, abrupt accelerations, and directional surges related to panic behavior. Sentiment-related features, which are based on the analysis of keywords and emotional tone, are obtained out of textual data related to social media, where specific words like "stuck," "pushed," or "scared" are used as indicators of a highly stressed crowd where the combination of both physical and emotional behavior of a crowd is considered.

C. Model Architecture and Training

YOLOv11 is used in real-time detection of people due to the high speed and accuracy of the algorithm when it comes to dense crowd settings. Each video frame is processed through the model to produce bounding boxes of individuals detected in the videos, out of which per-frame counts and space distributions of the crowd are obtained.

At the same time, the Lucas-Kanade optical flow algorithm is used to compute the velocity vectors and directional changes of single movements between the subsequent frames, and the combination of these operations aims to distinguish the common crowd movement from the abnormal rushing caused by panic. A surge is indicated by the change of the velocity of the crowd (ΔV) being more than 1.8 times the regular velocity flow (V) or the directional deviation of the flow ($\Delta\theta$) being more than 25.

Data is diverse and of varying quality, and the preprocessing has to ensure uniformity, correctness, and analysis readiness. Video Preprocessing: Different frames are resized to the same resolution to achieve the detection speed. The clarity can be enhanced by the noise-reduction filters, and normalization is used to eliminate the lighting contrast between daytime and nighttime. We use frame stabilization for optical flow to prevent camera shakes from influencing the movement detection. Facial Data Preprocessing: The facial regions of the video frames are cropped using the bounding boxes. This is further enhanced by applying normalization and correcting contrast for an even more precise capture of even subtle expressions in low lighting. The following training parameters are used for the ShanghaiTech dataset in order to translate dot annotations into bounding boxes for the model's training:

Epochs: 50
Batch Size: 16
Optimizer: AdamW
Learning Rate: 0.001

D. Visualization and Dashboard

Seaborn and Matplotlib in Python are used to display the detection and motion analysis results in the form of temporal heatmaps and density charts. Color coding is based on conventional risk, which means that red means high density and green and blue mean moderate and low density, respectively. The interactive analysis dashboard shows important real-time indicators such as maximum number of people, maximum density, maximum motion surge, and risk distribution over the entire monitored area, which helps the surveillance operators to identify the conditions of the crowd in real time and make decisions.

E. Alert and Response Mechanism

The system sounds alarms for security use by the dashboard once the crowd density or motion surge values surpass specified safety levels. Some preventive measures are entry gate control, crowd redirection guidance, and direct notification to on-ground authorities. The system allows more rapid and uniform intervention to enhance the overall outcomes of public safety in large-scale events by

complete automation of the response pipeline, which leads to reduced reliance on manual monitoring.

V. MODULES

A. Crowd Detection Module

This module is the start of the processing pipeline. Each preprocessed frame is applied to the YOLOv11 in real time to identify everybody who is in the scene. The model provides bounding boxes of every individual detected and the number of the people in a frame. YOLOv11 is chosen due to the fact that it can provide the correct predictions with low resolutions and has a low latency, which is suitable in live surveillance settings.

B. Crowd Behavior and Density Analysis Module

The detection output is subjected to two analyses by this module at the same time. A normal, crowded, or high-risk density map is generated per frame by the crowd density estimation element, which counts people in the frame and looks at the distribution of pixels. In order to determine the crowd's direction, velocity, and group movement patterns between frames, the Motion Tracking component employs the optical flow algorithm developed by Lucas-Kanade.

In Lucas-Kanade optical flow vectors, a surge is detected when there is an abrupt change in the speed or direction of travel. [11].

Mathematically:

- ΔV = change in average crowd velocity
- $\Delta\theta$ = change in movement direction variance
- $\tau_v = 1.8 \times \text{normal flow magnitude (velocity threshold)}$
- $\tau_\theta = 25^\circ$ (direction deviation threshold)

Condition:

If $\Delta V > \tau_v$ or $\Delta\theta > \tau_\theta$, the system flags a surge (abnormal crowd movement).

C. Visualization and Alert Module

The module translates processed outputs into operational and visual action responses. The output of density is plotted in color-coded heat maps with Seaborn and Matplotlib, with red cells signifying a high density of people and green or blue cells signifying a moderate or low congestion. An interactive surveillance board summarizes the number of people, movement trends, and status of threats to the security agents. When any high-risk event is triggered off, be it by the density classifier or the surge detector, the system automatically sends alerts to the security personnel, alerts authorities, opens and closes the entry gates, and directs crowds, which is a fully automated safety response pipeline.

VI. RESULTS AND DISCUSSION

The input image shows a highly crowded place. This was selected because these are the most probable situations where stampedes could occur. This is the evaluation metric table with the valuation scores. It describes the Precision, Recall and the mAP@0.5 score.

Table 1 - Evaluation metrics

Metrics	Score
mAP@0.5	0.83
Precision	0.89
recall	0.86

The first stage of processing is applying the YOLO v11 model to the frame. Once it is done, the individual persons are detected through bounding boxes produced by the YOLO v11.



Fig. 3. Original Image

Here, the bounding boxes are applied to the image as in Fig 3. From this, the crowd density estimation module finds the right middle or the left sections of the image. For instance, the right side has a very high density compared to the left side of the picture. The Lucas Kanadae method is used to track the movement of the crowd. This helps in finding the difference between a static crowd and a panic crowd (Fig 4).

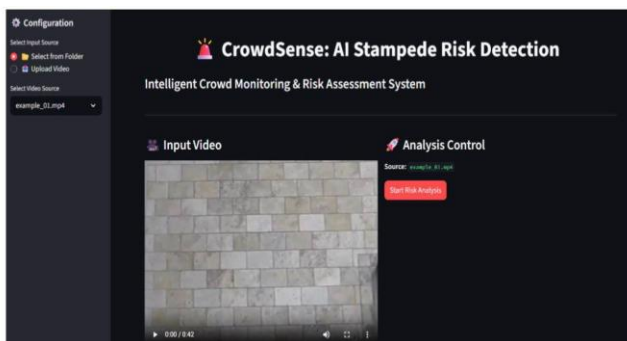


Fig. 4. User interface of the research

The heatmaps generated (temporal charts) in these images display varying colours in accordance with density. These are done using Python libraries like matplotlib and seaborn, which are very useful to precisely analyze the crowd density before moving into the next step as in Fig 5.

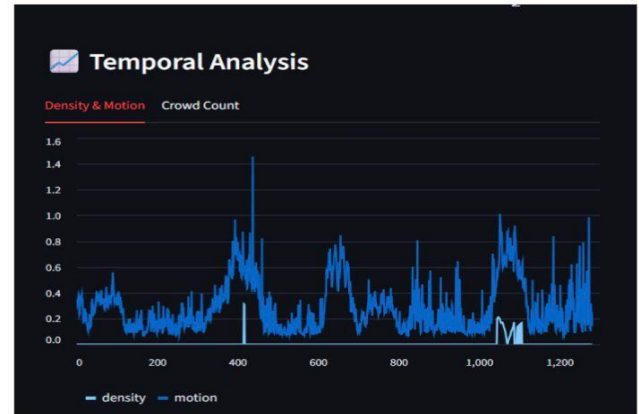


Fig. 5. Temporal analysis chart

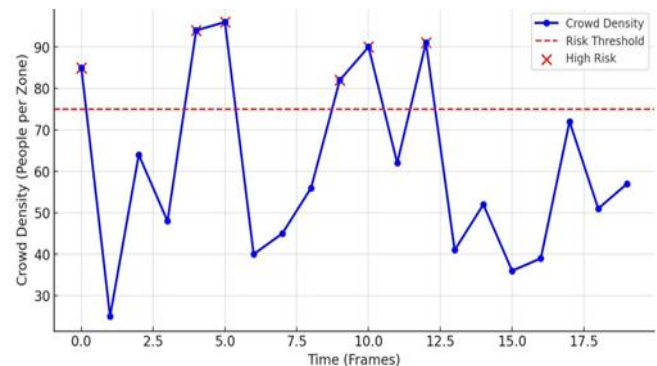


Fig. 6. Line chart for comparison

This line chart in figure 6, shows the comparison between the crowd density over time. As the red dotted arrow indicates, if the crowd exceeds a certain threshold, the risk factor of the stampede or similar events happening will be high.

This chart in figure 7, indicates the insights which has been so far derived from the chart, it shows the risk distribution measures, the max crowd count and the peak density of the crowd in addition to that it also shows the crowd surge value

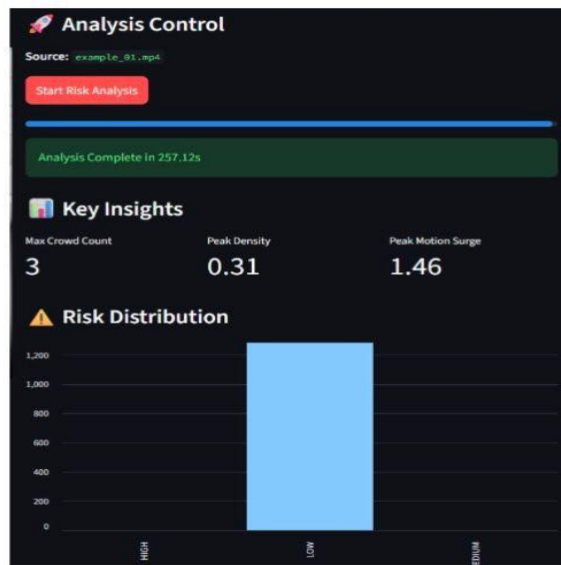


Fig.7 Analysis control chart

VII. CONCLUSION

This paper proposed an AI-based real-time crowd monitoring system that is based on YOLOv11 used in person detection, and the Lucas-Kanade optical flow technique used in motion tracking showed that the system is reliable and can accurately detect crowds in dense real-life conditions. The implemented CrowdSense dashboard gave spot views of crowd density, movement trends, and risk distribution to operators of the deployed surveillance and allowed intuitive evaluation of areas through heat maps on a temporal basis. The system has the benefit of being a full, end-to-end automation of the crowd surveillance system, with abnormal crowd surges automatically flagged and preventive measures activated against the crowd, including entry gate control and crowd redirection, presented as a scalable, cost-effective alternative to traditional manual surveillance. Future directions will include integration of multi-cameras, enhanced optical flow robustness in the high-density crowd, and adding social media sentiment analysis to the early warning system.

REFERENCES

- [1] Arefin, M., Wadud, M. A. H., & Rahman, A. (2024). Indoor and Outdoor Crowd Density Level Estimation with Video Analysis through Machine Learning Models. *arXiv preprint arXiv:2405.07419*.
- [2] Chaudhary, D., Kumar, S., & Dhaka, V. S. (2022). Video based human crowd analysis using machine learning: a survey. *Computer methods in biomechanics and biomedical engineering: imaging & visualization*, 10(2), 113-131.
- [3] Bhuiyan, M. R., Abdullah, J., Hashim, N., & Al Farid, F. (2022). Video analytics using deep learning for crowd analysis: a review. *Multimedia Tools and Applications*, 81(19), 27895-27922.

- [4] Vijay, K., Jeyanth, V., Kamalesh, S., & Dheva Prasath, D. (2024, November). Enhancing Operational Efficiency and Security in Multi-Sectioned Businesses with Edge AI through Real-Time Analytics. In *2024 International Conference on Electronic Systems and Intelligent Computing (ICESIC)* (pp. 251-256). IEEE.
- [5] Elbishlawi, S., Abdelpakey, M. H., Eltantawy, A., Shehata, M. S., & Mohamed, M. M. (2020). Deep learning-based crowd scene analysis survey. *Journal of Imaging*, 6(9), 95.
- [6] Jiang, R., Cai, Z., Wang, Z., Yang, C., Fan, Z., Chen, Q., ... & Shibasaki, R. (2021). DeepCrowd: A deep model for large-scale citywide crowd density and flow prediction. *IEEE Transactions on Knowledge and Data Engineering*, 35(1), 276-290.
- [7] Ma, X., Xi, W., Zhao, X., Chen, Z., Zhang, H., & Zhao, J. (2021). Wisual: Indoor crowd density estimation and distribution visualization using Wi-Fi. *IEEE Internet of Things Journal*, 9(12), 10077-10092.
- [8] Li, X., Chen, M., & Wang, Q. (2020). Quantifying and detecting collective motion in crowd scenes. *IEEE Transactions on Image Processing*, 29, 5571-5583.
- [9] Tahira, N. J., Suresh, P. D., & Park, J. S. (2024, October). Deep Learning based Approach for Crowd Density Estimation and Flow Prediction. In *2024 24th International Conference on Control, Automation and Systems (ICCAS)* (pp. 1274-1275). IEEE.
- [10] Zhang, X., Sun, Y., Li, Q., Li, X., & Shi, X. (2023). Crowd density estimation and mapping method based on surveillance video and GIS. *ISPRS International Journal of Geo-Information*, 12(2), 56.
- [11] Mansouri, W., Alohal, M. A., Alqahtani, H., Alruwais, N., Alshammeri, M., & Mahmud, A. (2025). Deep convolutional neural network-based enhanced crowd density monitoring for intelligent urban planning on smart cities. *Scientific Reports*, 15(1), 5759.
- [12] Deokate, S., Rajput, S., Nair, M., Parale, D., Mahamuni, A., & Nalla, P. (2024, December). Deep Learning-Based Crowd Surveillance and Density Estimation. In *2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAQSA)* (pp. 1-8). IEEE.
- [13] Reddy, M. J., Kowshik, A. C., Sharma, C. R., & Bhanushree, K. J. (2024, August). Crowd Control and Monitoring using Deep Learning. In *2024 Second International Conference on Networks, Multimedia and Information Technology (NMITCON)* (pp. 1-8). IEEE.
- [14] Abdullah, F., Abdelhaq, M., Alsaqour, R., Alatiyyah, M. H., Alnowaiser, K., Alotaibi, S. S., & Park, J. (2023). Context aware crowd tracking and anomaly detection via deep learning and social force model. *IEEE Access*, 11, 75884-75898.
- [15] Alia, A., Maree, M., Chraibi, M., Toma, A., & Seyfried, A. (2023). A cloud-based deep learning framework for early detection of pushing at crowded event entrances. *IEEE access*, 11, 45936-45949.